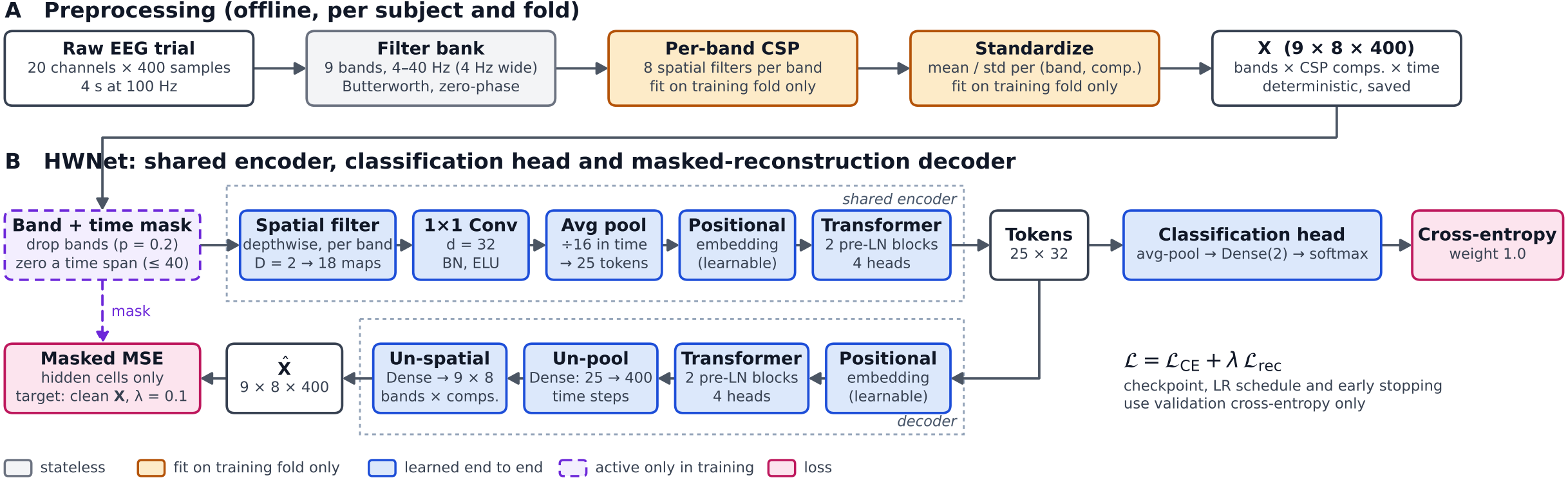
A Preprocessing (offline, per subject and fold)
Raw EEG trial
20 channels × 400 samples
4 s at 100 Hz
Filter bank
9 bands, 4–40 Hz (4 Hz wide)
Butterworth, zero-phase
Per-band CSP
8 spatial filters per band
fit on training fold only
Standardize
mean / std per (band, comp.)
fit on training fold only
X (9 × 8 × 400)
bands × CSP comps. × time
deterministic, saved
B HWNet: shared encoder, classification head and masked-reconstruction decoder
Band + time mask
drop bands (p = 0.2)
zero a time span (≤ 40)
mask
shared encoder
Spatial filter
depthwise, per band
D = 2 → 18 maps
1×1 Conv
d = 32
BN, ELU
Avg pool
÷16 in time
→ 25 tokens
Positional
embedding
(learnable)
Transformer
2 pre-LN blocks
4 heads
Tokens
25 × 32
Classification head
avg-pool → Dense(2) → softmax
Cross-entropy
weight 1.0
Masked MSE
hidden cells only
target: clean X, λ = 0.1
X̂
9 × 8 × 400
Un-spatial
Dense → 9 × 8
bands × comps.
Un-pool
Dense: 25 → 400
time steps
Transformer
2 pre-LN blocks
4 heads
Positional
embedding
(learnable)
decoder
$\mathcal{L} = \mathcal{L}_{CE} + \lambda \mathcal{L}_{rec}$
checkpoint, LR schedule and early stopping
use validation cross-entropy only
stateless
fit on training fold only
learned end to end
active only in training
loss